Proof Structure for the 3×3 Magic Square of Squares

Shaimaa Said Soltan

Abstract

This paper establishes a complete structural characterization of all 3×3 magic squares whose entries are perfect squares. By expressing the nine cells in terms of three parameters A^2, E^2, and X, we derive a universal system of equations governing every such square. These relations force each opposite pair to satisfy $U^2 + V^2 = 2E^2$, implying that the center E^2 is the average of all four opposite pairs. As a consequence, every row, column, and diagonal has magic sum $3E^2$, and all nine entries are determined algebraically by the three free parameters.

We show that requiring all nine entries to be perfect squares imposes the additional constraint $A^2 + B^2 = 2E^2$, which forces the bottom-left cell X to equal the center E^2. This produces a repeated value in the square, and therefore no 3×3 magic square of nine distinct perfect squares can exist. The only possible solutions are those in which exactly two of the nine entries are not squares.

This structural result explains the well-known 425^2 magic square of Bremner and Sallows, which contains exactly seven perfect squares and two non-square entries. Its layout matches the universal equations precisely, and its two non-square cells occur exactly where the theory predicts they must. Thus the 425^2 example is not an isolated curiosity but the canonical representative of all possible seven-square magic squares. The analysis presented here proves that seven distinct squares is the maximal achievable number in a 3×3 magic square.

Magic Square Setup

We label the 3×3 magic square as:

$$\begin{array}{ccc} A^2 & B^2 & C^2 \\ D^2 & E^2 & F^2 \\ X & H^2 & I^2 \end{array}$$

where the center is E^2 and the cell G is denoted by X.

Defining Equations

The magic square structure imposes the following six relations:

$$\begin{aligned}
&(1) \quad D^2 = 3E^2 - A^2 - X, && (1)\\
&(2) \quad F^2 = A^2 + X - E^2, && (2)\\
&(3) \quad I^2 = 2E^2 - A^2, && (3)\\
&(4) \quad H^2 = 3E^2 - X - I^2, && (4)\\
&(5) \quad C^2 = 3E^2 - E^2 - X = 2E^2 - X, && (5)\\
&(6) \quad B^2 = 3E^2 - A^2 - C^2 = E^2 - A^2 + X. && (6)
\end{aligned}$$

Opposite Cell Sums

Opposite cells in any 3×3 magic square must satisfy:

$$\text{(opposite pair)} = 2E^2.$$

Using the equations above:

1. $D^2 + F^2 = 2E^2$

$$D^2 + F^2 = (3E^2 - A^2 - X) + (A^2 + X - E^2) = 2E^2.$$

2. $B^2 + H^2 = 2E^2$

Since

$$H^2 = 3E^2 - X - (2E^2 - A^2) = E^2 - X + A^2,$$

we have

$$B^2 + H^2 = (E^2 - A^2 + X) + (E^2 - X + A^2) = 2E^2.$$

3. $C^2 + X = 2E^2$

$$C^2 + X = (2E^2 - X) + X = 2E^2.$$

4. $A^2 + I^2 = 2E^2$

$$A^2 + I^2 = A^2 + (2E^2 - A^2) = 2E^2.$$

Thus all opposite pairs satisfy the magic square identity.

Row, Column, and Diagonal Sums

Since each opposite pair sums to $2E^2$ and the center is E^2, every row, column, and diagonal has total:

$$\text{Line Sum} = E^2 + 2E^2 = 3E^2.$$

Thus the magic square structure is preserved.

Closing the Square: The Condition on X

From equation (6),

$$B^2 = E^2 - A^2 + X.$$

But the magic square also requires:

$$A^2 + B^2 = 2E^2.$$

Substituting B^2:

$$A^2 + (E^2 - A^2 + X) = 2E^2,$$

which simplifies to:

$$E^2 + X = 2E^2,$$

hence

$$X = E^2.$$

Conclusion

To guarantee that B^2 is a perfect square within this universal magic-square structure, one must set:

$$X = E^2.$$

This forces repetition of the value E^2 in the square, proving that:

No 3×3 magic square of nine distinct perfect squares can exist.

Example: The 425^2 Magic Square with Seven Squares

A celebrated example of a 3×3 magic square containing exactly seven perfect squares, discovered by Bremner and Sallows, is:

$$\begin{array}{ccc} 205^2 & 360721 & 373^2 \\ 527^2 & 425^2 & 289^2 \\ 222121 & 23^2 & 565^2 \end{array}$$

In this arrangement:

- The center cell is $E^2 = 425^2$.
- Seven entries are perfect squares: $205^2,\ 373^2,\ 527^2,\ 425^2,\ 289^2,\ 23^2,\ 565^2$.
- Two entries, 360721 and 222121, are not squares.

All rows, columns, and diagonals sum to the same magic total M:

$$205^2 + 360721 + 373^2 = 527^2 + 425^2 + 289^2 = 222121 + 23^2 + 565^2 = M.$$

This example is important because its layout matches the universal equation system developed earlier:

$$\begin{matrix} A^2 & B^2 & C^2 \\ D^2 & E^2 & F^2 \\ X & H^2 & I^2 \end{matrix}$$

Here the bottom-left cell is X, and the top-middle cell is B^2, exactly as in the algebraic setup. Thus the closing of the nine-cell structure depends on the conditions that X and B must both be perfect squares.

The universal equations proved in this paper show that the only way to force B^2 to be square is to set

$$X = E^2,$$

which necessarily repeats the value E^2 in the square. Therefore, while a 3×3 magic square can contain seven perfect squares, it cannot contain nine distinct perfect squares.

Comparison Between the Universal Equations and the 425^2 Example

The universal 3×3 magic square of squares developed earlier is:

$$\begin{matrix} A^2 & B^2 & C^2 \\ D^2 & E^2 & F^2 \\ X & H^2 & I^2 \end{matrix}$$

with the structural equations:

$$\begin{aligned} D^2 &= 3E^2 - A^2 - X, \\ F^2 &= A^2 + X - E^2, \\ I^2 &= 2E^2 - A^2, \\ H^2 &= 3E^2 - X - I^2, \\ C^2 &= 2E^2 - X, \\ B^2 &= E^2 - A^2 + X. \end{aligned}$$

A known magic square containing seven perfect squares, with center $E^2 = 425^2$, is:

$$\begin{array}{ccc} 205^2 & 360721 & 373^2 \\ 527^2 & 425^2 & 289^2 \\ 222121 & 23^2 & 565^2 \end{array}$$

We now compare each cell of this example to the universal form.

Mapping of Cells

$$\begin{aligned} A^2 &= 205^2, \\ B^2 &= 360721, \\ C^2 &= 373^2, \\ D^2 &= 527^2, \\ E^2 &= 425^2, \\ F^2 &= 289^2, \\ X &= 222121, \\ H^2 &= 23^2, \\ I^2 &= 565^2. \end{aligned}$$

Verification of Opposite-Pair Equations

Using the universal identities:

$$A^2 + I^2 = 205^2 + 565^2 = 2E^2,$$

$$B^2 + H^2 = 360721 + 23^2 = 2E^2,$$

$$C^2 + X = 373^2 + 222121 = 2E^2,$$

$$D^2 + F^2 = 527^2 + 289^2 = 2E^2.$$

Thus the 425^2 example satisfies *all* four opposite-pair equations required by the universal structure.

Verification of Row, Column, and Diagonal Sums

Each line contains the center E^2 plus one opposite pair:

$$\text{Line Sum} = E^2 + 2E^2 = 3E^2,$$

which matches the example:

$$205^2 + 360721 + 373^2 = 527^2 + 425^2 + 289^2 = 222121 + 23^2 + 565^2 = 3E^2.$$

Why This Example Has Only Seven Squares

In the universal equations, the top-middle cell is:

$$B^2 = E^2 - A^2 + X.$$

To guarantee B^2 is a perfect square for *all* valid magic squares, the structure forces:

$$A^2 + B^2 = 2E^2 \quad \Rightarrow \quad X = E^2.$$

This repeats the center value, making a nine-square magic square impossible.

In the 425^2 example, $X = 222121$ is *not* a square, so $B^2 = 360721$ is also not a square. This avoids the forced repetition $X = E^2$ and allows the square to exist with exactly seven perfect squares.

Conclusion

The 425^2 magic square fits the universal equations perfectly, but it avoids the forced condition

$$X = E^2$$

by allowing two non-square entries. Your universal structure therefore explains exactly why:

Seven squares are possible, but nine distinct squares are impossible.

Theorem: Impossibility of a 3×3 Magic Square of Nine Distinct Squares

Theorem. Let

$$\begin{matrix} A^2 & B^2 & C^2 \\ D^2 & E^2 & F^2 \\ X & H^2 & I^2 \end{matrix}$$

be any 3×3 magic square whose entries are perfect squares and whose center is E^2. Then the universal magic-square relations

$$\begin{aligned} D^2 &= 3E^2 - A^2 - X, \\ F^2 &= A^2 + X - E^2, \\ I^2 &= 2E^2 - A^2, \\ H^2 &= 3E^2 - X - I^2, \\ C^2 &= 2E^2 - X, \\ B^2 &= E^2 - A^2 + X, \end{aligned}$$

hold for every such square.

If all nine entries are required to be perfect squares, then the magic-square condition

$$A^2 + B^2 = 2E^2$$

forces

$$B^2 = E^2 - A^2 + X,$$

and hence

$$E^2 + X = 2E^2,$$

which implies

$$X = E^2.$$

Therefore the bottom-left cell X must equal the center E^2, producing a repeated value in the square. Consequently, a 3×3 magic square with nine *distinct* perfect squares cannot exist.

Corollary. The maximum number of distinct perfect squares that can appear in a 3×3 magic square is seven, as demonstrated by the known 425^2 example.

Final Summary: Connection to the 425^2 Example

The classical magic square with center 425^2,

$$\begin{array}{ccc} 205^2 & 360721 & 373^2 \\ 527^2 & 425^2 & 289^2 \\ 222121 & 23^2 & 565^2 \end{array}$$

is a perfect illustration of the universal structure.

- It satisfies all opposite-pair identities:

$$205^2 + 565^2 = 2 \cdot 425^2, \qquad 360721 + 23^2 = 2 \cdot 425^2,$$

$$373^2 + 222121 = 2 \cdot 425^2, \qquad 527^2 + 289^2 = 2 \cdot 425^2.$$

- It satisfies all six universal equations relating A^2, E^2, X to the remaining cells.
- It contains exactly seven perfect squares, with the two non-square entries appearing precisely where the universal equations allow them: in B^2 and X.

If one attempts to make B^2 a perfect square, the universal identity

$$A^2 + B^2 = 2E^2$$

forces

$$X = E^2,$$

which repeats the center value and destroys the distinctness of the nine entries.

Thus the 425^2 example is not an isolated curiosity: it is the *canonical* representative of all possible seven-square magic squares, and it fits perfectly into the algebraic framework developed in this paper.

Seven squares are possible; nine distinct squares are impossible.

Structural Constraints on the Squareness of X and B

The universal equations for a 3×3 magic square of squares are:

$$B^2 = E^2 - A^2 + X, \qquad A^2 + B^2 = 2E^2.$$

These two relations determine the interaction between the squareness of the bottom-left cell X and the top-middle cell B^2.

Case 1: X square, B not square (structurally possible)

Suppose X is a perfect square. Then $B^2 = E^2 - A^2 + X$ is determined algebraically, but there is no requirement that B^2 itself be a perfect square. The magic-square condition

$$A^2 + B^2 = 2E^2$$

still holds automatically, and no contradiction arises.

Thus it is structurally possible to have:

$$X \text{ square}, \qquad B \text{ not square}.$$

This configuration corresponds to a hypothetical magic square with eight perfect squares (only B non-square), and the universal equations do not forbid such a structure.

Case 2: B square, X not square (structurally impossible)

Now suppose B^2 is required to be a perfect square. Using the universal identity

$$A^2 + B^2 = 2E^2,$$

and substituting $B^2 = E^2 - A^2 + X$, we obtain:

$$A^2 + (E^2 - A^2 + X) = 2E^2,$$

which simplifies to:

$$E^2 + X = 2E^2, \qquad \Rightarrow \qquad X = E^2.$$

Therefore, if B^2 is a perfect square, then X must equal the center E^2, and hence X is also a perfect square. This produces a repeated value in the magic square.

Thus it is structurally impossible to have:

$$B \text{ square}, \qquad X \text{ not square}.$$

Conclusion

The universal equations allow X to be square without forcing B to be square, but they do not allow B to be square without forcing X to equal E^2. This asymmetry explains why seven-square magic squares exist, while nine-square magic squares do not.

1 Relations Satisfied by B^2

Since $B^2 = E^2 - A^2 + X$ is determined by the three free parameters, it may be expressed in terms of each remaining cell:

$$B^2 = E^2 - A^2 + X \tag{7}$$
$$B^2 = 3E^2 - A^2 - C^2 \tag{8}$$
$$B^2 = E^2 - C^2 + I^2 \tag{9}$$
$$B^2 = 2I^2 - D^2 \tag{10}$$
$$B^2 = 2E^2 - 2A^2 + F^2 \tag{11}$$
$$B^2 = -A^2 + F^2 + I^2 \tag{12}$$
$$B^2 = 2X - F^2 \tag{13}$$
$$B^2 = 2E^2 - H^2 \tag{14}$$

Writing these as a coefficient matrix over the ordered basis $(E^2, A^2, C^2, D^2, F^2, X, H^2, I^2)$:

$$M = \begin{pmatrix} 1 & -1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 3 & -1 & -1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & -1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 2 \\ 2 & -2 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 2 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix}$$

Every row of M sums to 1. This is forced by the degenerate magic square in which all nine entries are equal: substituting it into any valid relation gives $t = \left(\sum_j m_{ij}\right) t$, so the row sums serve as a consistency check on the derivations.

Each relation holds identically on the three-parameter family. Two are of particular arithmetic interest:

$$B^2 + F^2 = 2X, \qquad B^2 + D^2 = 2I^2,$$

placing B^2 at the end of two distinct three-term arithmetic progressions. If B, D, F, I and X are all squares, these

2 The Parameter Triple (E^2, I^2, X)

The three opposite-pair identities may be written as

$$B^2 + D^2 = 2I^2, \tag{15}$$
$$B^2 + F^2 = 2X, \tag{16}$$
$$D^2 + F^2 = 2E^2. \tag{17}$$

Thus I^2, X and E^2 are the midpoints of the three pairs formed from B^2, D^2 and F^2.

Subtracting (**??**) from (**??**) eliminates B^2:

$$D^2 - F^2 = 2\,(I^2 - X).$$

Adding this to (**??**) gives $2D^2 = 2E^2 + 2(I^2 - X)$, and subtracting it gives $2F^2 = 2E^2 - 2(I^2 - X)$. Substituting either result into (**??**) yields B^2. Hence

$$B^2 = -E^2 + I^2 + X, \tag{18}$$
$$D^2 = \quad E^2 + I^2 - X, \tag{19}$$
$$F^2 = \quad E^2 - I^2 + X. \tag{20}$$

The three expressions differ only by the placement of the minus sign, and each pairwise sum recovers the corresponding identity: $B^2 + D^2 = 2I^2$, $B^2 + F^2 = 2X$, $D^2 + F^2 = 2E^2$. Consequently (E^2, I^2, X) may be taken as the free parameters of the family in place of (A^2, E^2, X), with $A^2 = 2E^2 - I^2$, $C^2 = 2E^2 - X$ and $H^2 = 2E^2 - B^2$ recovering the remaining cells.